4.1.2. Concatenate Operation on Single Linked List

Code :

```
struct node {
   int data;
   struct node* next;
};

typedef struct node* NODE;

NODE createNode(int x) {
 NODE newNode = (NODE)malloc(sizeof(struct node));
   if (newNode == NULL) {
      printf("Memory allocation failed!\n");
      return NULL;
   }
   newNode->data = x;
   newNode->next = NULL;
   return newNode;









}

NODE addNode(NODE first, int x) {
 NODE newNode = createNode(x);
   if (first == NULL) {
```

```
        return newNode;
    }

    NODE temp = first;
    while (temp->next != NULL) {
        temp = temp->next;
    }
    temp->next = newNode;
    return first;




}

NODE concatenate(NODE t1, NODE t2) {
 if (t1 == NULL) return t2;
    if (t2 == NULL) return t1;

    NODE temp = t1;

    while (temp->next != NULL) {
        temp = temp->next;
    }

    temp->next = t2;

    return t1;
```

```
}

void displayList(NODE first) {
 if (first == NULL) {
      printf("NULL\n");
      return;
   }

   NODE temp = first;
   while (temp != NULL) {
      printf("%d --> ", temp->data);
      temp = temp->next;
   }
   printf("NULL\n");





}
```

Programme :

CODETANTRA Home

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4.1.2. Concatenate Operation on Single Linked List

07:04

Write a C program to implement a menu-driven application for performing operations on two Singly Linked Lists. The program should allow the user to create two singly linked lists, display them, and concatenate the second list to the end of the first list.

Functions need to be implemented in **AllOperations.c:**

1. **NODE createNode(int x)**
   - Create a new node using dynamic memory allocation, assign the given value to the data field, initialize the next pointer to NULL, and return the address of the newly created node.
2. **NODE addNode(NODE first, int x)**
   - Insert a new node with the given value at the end of the specified linked list and return the updated head pointer.
3. **NODE concatenate(NODE t1, NODE t2)**
   - Concatenate the second singly linked list to the end of the first singly linked list and return the head pointer of the resulting list.
   - If either list is empty, return the non-empty list.
4. **void displayList(NODE first)**
   - Traverse and display the elements of the linked list in the format:

```
data --> data --> ... --> NULL
```

Menu Options:

The program should repeatedly display the following menu until user selects exit operation:

```
1. Create SLL1
2. Create SLL2
3. Display SLL1
```

Explorer

AllOperat...

Submit

```
    return t1;


}

void displayList(NODE first) {
 if (first == NULL) {
        printf("NULL\n");
```

Average time 0.079 s 78.67 ms

Maximum time 0.085 s 85.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 84 ms Debug

Expected output	Actual output
1. Create SLL1	1. Create SLL1
2. Create SLL2	2. Create SLL2
3. Display SLL1	3. Display SLL1
4. Display SLL2	4. Display SLL2
5. Concatenation	5. Concatenation
6. Exit	6. Exit

Terminal Test cases

Debugger

< Prev Reset Submit Next >